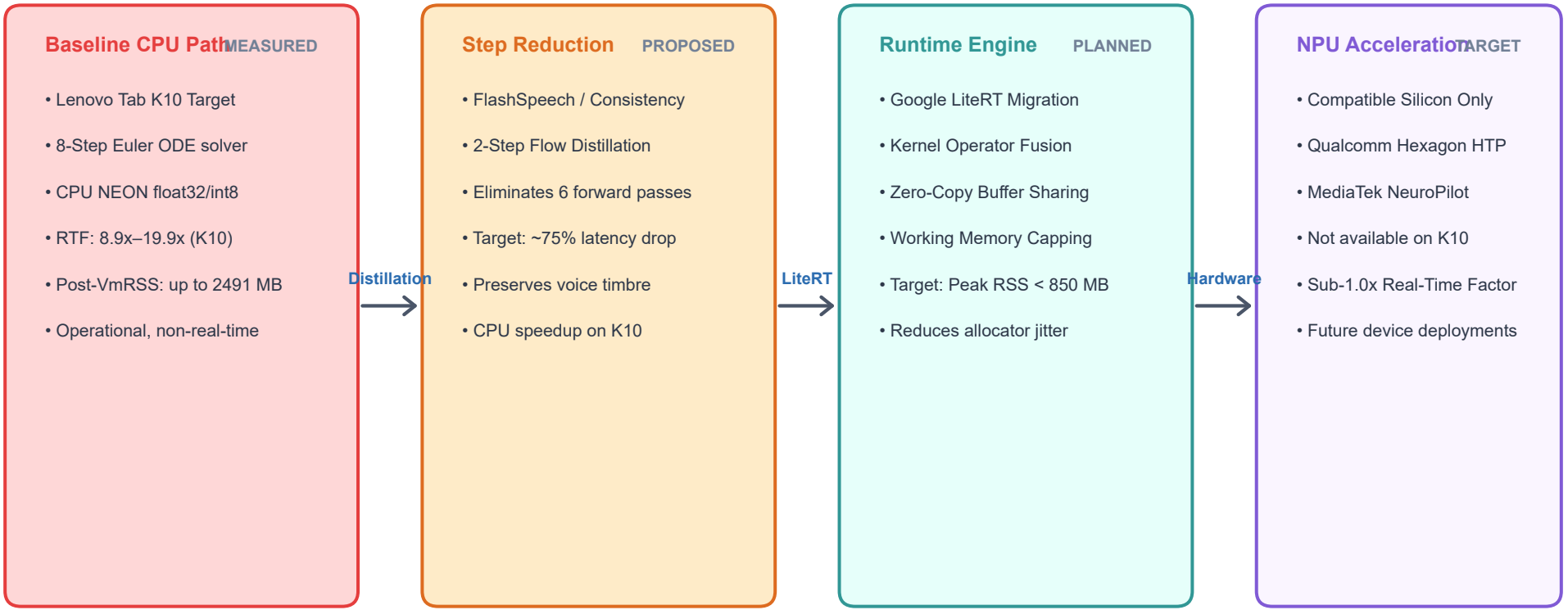

Baseline CPU Path
MEASURED
• Lenovo Tab K10 Target
• 8-Step Euler ODE solver
• CPU NEON float32/int8
• RTF: 8.9x–19.9x (K10)
• Post-VmRSS: up to 2491 MB
• Operational, non-real-time
Distillation
Step Reduction
PROPOSED
• FlashSpeech / Consistency
• 2-Step Flow Distillation
• Eliminates 6 forward passes
• Target: ~75% latency drop
• Preserves voice timbre
• CPU speedup on K10
LiteRT
Runtime Engine
PLANNED
• Google LiteRT Migration
• Kernel Operator Fusion
• Zero-Copy Buffer Sharing
• Working Memory Capping
• Target: Peak RSS < 850 MB
• Reduces allocator jitter
Hardware
NPU Acceleration
TARGET
• Compatible Silicon Only
• Qualcomm Hexagon HTP
• MediaTek NeuroPilot
• Not available on K10
• Sub-1.0x Real-Time Factor
• Future device deployments